CS 4375
ASSIGNMENT 2

Names of students in your group: Kylie Quinney (krq210000)

Number of free late days used: 0

Note: You are allowed a **total** of 4 free late days for the **entire semester**. You can use at most 2 for each assignment. After that, there will be a penalty of 10% for each late day.

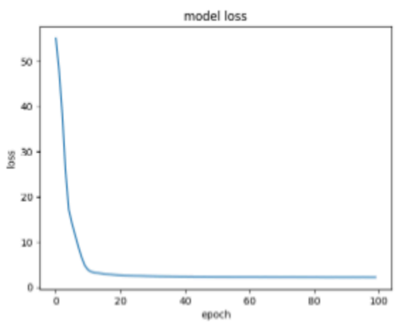
Please list clearly all the sources/references that you have used in this assignment.

Table demonstrating hyper-pramaters and evaluation metrics for each model version:

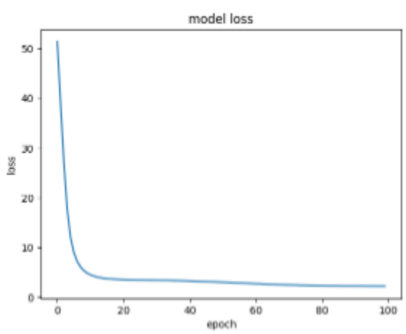
Model #	activation	learning_rate	max_iterations	hidden_layer_sizes	Train mse	Train mae	Train accuracy	Test mse	Test mae	Test accuracy
1	relu	constant	100	30,30,30	4.265357822	1.451284236	0.5898101959	4.407986009	1.465586409	0.5750768944
2	tanh	constant	100	30,30,30	4.23846298	1.469911427	0.589870816	4.483992152	1.503713885	0.5748493732
3	tanh	constant	100	100,100	4.312935064	1.493125346	0.5781431722	4.497316908	1.504607511	0.5871293994
4	tanh	constant	200	100,100	3.981007644	1.440359839	0.6004521235	4.738508334	1.549013673	0.5941677674
5	tanh	adaptive	300	100,100,100	4.250430712	1.509762915	0.5899467597	4.155057512	1.472818058	0.6032016977
6	tanh	adaptive	400	100,100,100	4.11840704	1.46102219	0.5899259161	4.379746803	1.471813633	0.6161957938

Model history plots for each of the models listed above:

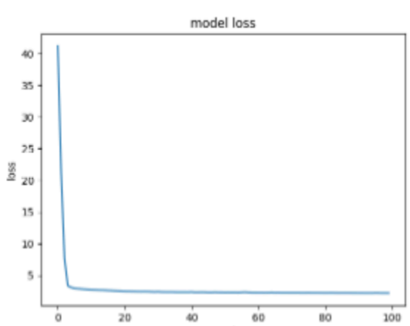
Model #1:



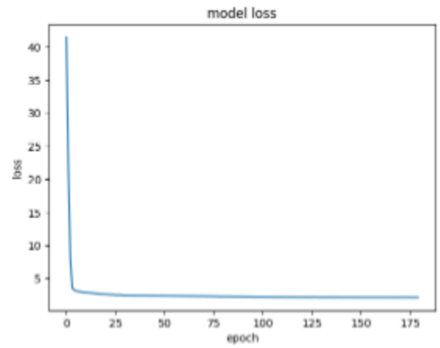
Model #2:



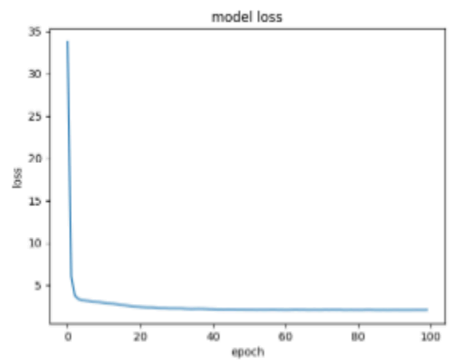
Model #3:



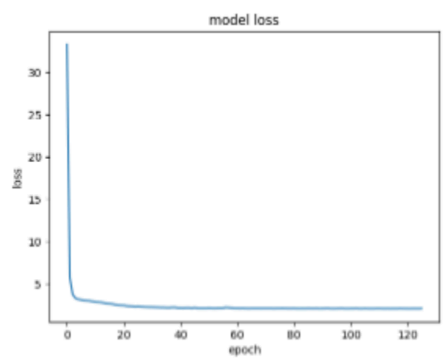
Model #4:



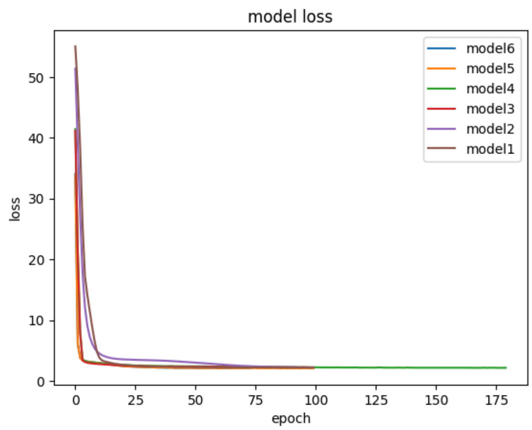
Model #5:



Model #6:



Model history plot for all of the models listed above:



Summary of results:

For my dataset, the tanh activation function worked the best. After reading up on relu v tanh activation, it seems that most people consider relu to be better, yet tanh worked better for me, so I was left wondering why this is the case. After some further investigation, it seems that tanh can work better than relu if your model's hyper-parameters are set up to optimize the tanh function rather than the relu. It is possible that relu could have gotten me even better results if I would have spent a few more hours tinkering with the hyperparameters while using relu.

My model did tend to work better with higher epochs and larger sized hidden layers, however this was only the case up to a certain point. After 400 epochs the model would overfit, causing my training score to go up, but my testing score to go down. This was also the case when trying to add more hidden layers and/or larger hidden layers.

Link to data used:
https://raw.githubusercontent.com/kyliequinney/WineQualityData/refs/heads/main/abalone.csv

Sources:
https://www.aitude.com/comparison-of-sigmoid-tanh-and-relu-activation-functions/